

Sara Rostami

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SUMMARY

ML researcher with 4+ years taking messy, high-dimensional problems from raw data to working systems, now focused on generative and agentic AI — RAG, LLM agents, fine-tuning, and evaluation. Prior MSc in Artificial Intelligence and Robotics. Seeking Applied Scientist, AI/ML Engineer, and Data Scientist roles.

TECHNICAL SKILLS

GenAI & Agentic AI: LangChain, LangGraph, Hugging Face, OpenAI / Gemini APIs — RAG, autonomous agents, tool integration, prompt engineering, fine-tuning & domain adaptation, model evaluation

ML & Deep Learning: PyTorch, Scikit-learn, Keras — trustworthy AI, explainable AI (SHAP, LIME, D-RISE), uncertainty quantification & calibration, time-series forecasting, multivariate decoding (MVPA)

Languages: Python, SQL, R, MATLAB, Java, HTML/CSS

Deployment & Data: FastAPI, Git, PostgreSQL, SQL Server, pandas, NumPy, SciPy, Selenium, BeautifulSoup — Matplotlib, Seaborn, Plotly

SELECTED PROJECTS

Calibrated Uncertainty in Vision-Language Models — Benchmarked post-hoc calibration methods for a BLIP-2 vision-language model on VQA v2: temperature scaling cut expected calibration error from 0.321 to 0.097, outperforming a Laplace approximation. [GitHub](#)

Neutro-Sentry — Hack Western 12 — Multimodal stroke-risk triage tool built end-to-end in a 36-hour hackathon. A SwiftUI iOS client streams facial-mesh landmarks and vitals to a Python/FastAPI backend, where a Gemini pipeline grounded in clinical guideline documents returns structured risk scores and explanations to a React/TypeScript dashboard. [Devpost](#)

Explainable AI: SHAP, LIME, and D-RISE — SHAP (Deep/Kernel) on a WHO life-expectancy network identifying schooling as the dominant predictor; D-RISE for model-agnostic object-detection explanations; LIME superpixel saliency on MobileNetV2. [GitHub](#)

Additional projects — BERT & BEiT attention implementations from first principles, GANs (DCGAN/AC-GAN/WGAN-GP), fraud detection and OCR with deep networks, cohort-based customer retention analysis, audio classification. github.com/SaraRostami/University

EXPERIENCE

Graduate Researcher — Mohsenzadeh Lab

Sep 2025 – Present

University of Western Ontario & Vector Institute — London, ON, Canada

- Designed and run a simultaneous EEG/fNIRS acquisition protocol — 32-channel EEG co-registered with 32 fNIRS optodes — recording 20+ participants across 5 image categories.
- Built the study's end-to-end preprocessing, artifact-rejection, and feature-extraction pipeline, replacing a manual workflow with one reproducible run per subject across all 20+ datasets.
- Ran multivariate decoding over 936 trials per participant with permutation testing, linking neural signals to behavioural measures of visual category perception and image memorability.
- Benchmarking learned representations from vision models (AlexNet, ResNet, and larger pretrained backbones) against the recorded neural data to isolate which features predict memorability — with downstream applications in designing visual content that is measurably more memorable, such as marketing and advertising material.

Research Assistant — Convergent Technologies Research Center

Aug 2022 – Aug 2025

University of Tehran — Tehran, Iran

- Acquired and preprocessed 128-channel EEG from 15 participants; applied MVPA to identify cognitive correlates of repetition and expectation effects in visual mismatch negativity.

Research Intern — Genzel Lab, Donders Institute

Aug 2022 – Feb 2023

Radboud University — Nijmegen, Netherlands

- Compared 4 classifiers on physiological sleep recordings; used random-forest feature importance to isolate the top predictors of memory consolidation.

EDUCATION

MSc Neuroscience (Machine Learning Specialization), University of Western Ontario — GPA 3.97/4.0

Sep 2025 – Aug 2027

MSc Artificial Intelligence and Robotics, University of Tehran — GPA 19.12/20 ($\approx 3.9/4.0$), ranked 3rd; Trustworthy AI coursework

2021 – 2024

BSc Computer Engineering, Babol Noshirvani University of Technology

2016 – 2021

AWARDS & TEACHING

Funding & awards: Vector Institute Research Grant (2025–2026) · Western University Graduate Funding Package (2025–2027).

Teaching: Neuromatch Academy Comp. Neuroscience Summer School (2023, 2025, 2026), project-pod TA; Western University (2026), Python and Java; University of Tehran (2023), Statistical Inference, Cognitive Neuroscience.